

Checkpoint S3 — CTAF 2021 Annual Report

Consolidated Extraction: Case Studies + Structural Data

Real-Time Anomaly Detection Platform — Amen Bank

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June 2026

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1 Source Identification

Field	Value
Source ID	S3
Document	Rapport Annuel 2021, Commission Tunisienne des Analyses Financières
Authority	CTAF (Tunisian FIU, Egmont Group member)
Language	French
Pages	70
Period covered	Calendar year 2021 (statistics cut 15 Jan 2022)
Extraction method	Markdown OCR + manual verification
Extraction template	v2 (3 patches + 9th coverage dimension)

Positioning in calibration strategy. The Amen Bank 2023 Annual Report (S2) calibrates the *normal population* — the 99%+ of legitimate operations. CTAF calibrates the *flagged population* — the <1% that should trigger alerts. Both are required inputs to the synthetic data generator.

Known limitation. CTAF data reflects *banks' notification behaviour*, not underlying criminal behaviour. The distributions extracted here are filtered through whatever detection systems existed in 2021. This is a known limitation of typology-driven anomaly modelling and must be acknowledged in the final rapport.

2 Trust Profile (v2 — 7 Dimensions)

Dimension		Score	Justification
Recency		Strong	2021 data, published 2022
Authority		Strong	National FIU, statutory mandate under Loi organique 2015-26
Peer Review		Medium	GAFI/GAFIMOAN mutual evaluation framework, but this specific report is self-published
Information Volume		Strong	1,538 dossiers treated, 1,209 disseminated; 20,222 BBE operations
Geographic	Relevance	Strong	Tunisian financial system, directly covers Amen Bank's regulatory environment
Data Type		Medium	Aggregate statistics + 4 anonymised case studies; no raw transaction data
Reproducibility		Weak	Anonymised cases, aggregate percentages; underlying data not accessible
Tally: 4 Strong / 2 Medium / 1 Weak			

3 Coverage Matrix (v2 — 9 Dimensions)

Dimension	Coverage	Notes
Amount distribution	High	Financial tier table (6 tiers, 5-year series); BBE concentration table (8 intervals)

Dimension	Coverage	Notes
Operation type mix	High	Instrument co-occurrence rates (68/56/49/34/13%); BBE structure by op type
Transaction frequency	Medium	Recurrence tables for BBE; passeur threshold (5+ ops)
Inter-event time	Low	Slow-drip 10-year pattern (Case 1); no general inter-event stats
Temporal patterns	Low	Quarterly BBE peaks (Q3); no intra-day or day-of-week data
Spatial / branch	Medium	Geographic cartography across 24 governorates; top-8 concentration
Channel mix	Not covered	No channel-level breakdown in CTAF reporting
Beneficiary / emitter known-rate	Medium	Known beneficiary tracking in Cases 1, 2, 4; passeur analysis
Op-sequence / transition	High	All 4 cases describe multi-op sequences; co-occurrence rates confirm multi-instrument schemes

4 Anomaly Typology

4.1 Behavioural Shapes (v2 §3a)

1. **Slow-drip third-party cash accumulation** (Case 1) — repeated small versements by third parties over years, full cash conversion via retraits
2. **Multi-stage placement–layering–integration** (Case 2) — foreign wire fan-in, product cycling, asset acquisition
3. **PPE-to-PPE corporate–personal flow** (Case 3) — round-amount chèques just under threshold, immediate cash disposal, reverse flow
4. **KYC inconsistency + commercial conduit via personal account** (Case 4) — undeclared business role, personal EUR account used for supplier payments, multi-physical-person fan-in
5. **Threshold structuring / smurfing** (BBE structural data) — 85% of BBE operations cluster in 5–20k TND band around regulatory threshold
6. **Passeur de fonds pattern** (BBE versement analysis) — occasional clients making 5+ versements for same beneficiary

4.2 Institutional Framing (v2 §3b)

Category	CTAF split	Our scope
Blanchiment d’argent (BA)	85% of transmitted dossiers	Primary
Financement du terrorisme (FT)	15% of transmitted dossiers	Secondary

Predicate offences (2021 transmitted dossiers):

Infraction sous-jacente	%
Contrebande	20%
Corruption et trafic d'influence	19%
Escroquerie & Fraude	11%
Faux et usage de faux	9%
Other (long tail: 8 categories)	41%

Generator use: predicate taxonomy drives 3–4 distinct anomaly injection scenario families (cash-heavy $\mapsto$ contrebande, transfer-based $\mapsto$ corruption, document-based $\mapsto$ escroquerie/faux), weighted approximately 20/19/11.

5 Case Study Summaries

5.1 Case 1 — Terrorism Financing (p. 15–17)

Actors	X (Tunisian, imam, multi-role: gérant jardin d'enfants + société de commerce + association founder). Third-party depositors Y, Z (linked to extremist orgs). Foreign transferor from XLand.
Pattern	Slow drip over 10 years: third-party cash deposits (84,630 TND cumulative), sporadic foreign incoming wires with implausible motive (“aide familiale”), full cash conversion. Profile–flux inconsistency.
Predicate	Financement du terrorisme
CTAF outcome	CNLCT listing (sanctions-equivalent) + transmission to Parquet
Key signals	<code>is_third_party_deposit</code> , <code>lifetime_cash_conversion_ratio</code> $\approx$ 1.0, <code>is_foreign_inbound_implausible_motive</code> , archetype z-score deviation
Schema impact	<code>depositor_id</code> on versement payload confirmed load-bearing; <code>lifetime_*</code> aggregates at client level (not per-account)

5.2 Case 2 — Money Mule Network (p. 18–19)

Actors	X (Franco-Tunisian-Swiss-British, non-resident) and Y (Anglo-Lebanese), married couple in L-Land. Multiple shell companies (1 GBP capital).
Pattern	Three classic laundering stages: placement (foreign incoming wires, 3.5 MD enjeu), layering (assurance-vie 2.9 MD broken after 1 year, bons de caisse 2.4 MD cycled every 3 months), integration (clinic shares, real estate, vehicle). Multi-account, multi-currency.
Predicate	Fraud + corruption
CTAF outcome	Asset freeze + transmission to Procureur + judicial inquiry
Scope note	Layering/integration phases (investment products) are out of scope for our 4-op system. System catches placement-phase signals only. Documented as honest scope limitation.
Schema impact	<code>transit_velocity_*</code> , <code>multi_currency_activity_ratio_*</code> , <code>cross_jurisdiction_outflow_distribution</code> . <code>sector_code</code> added to <code>accounts</code> table (D14).

5.3 Case 3 — Corruption + Environmental Crime (p. 20–21)

Actors	X (PPE, g�rant ALPHA — scrap metal recycling), Y (PPE, dismissed administrative official), Z (straw man, fictitious real estate cover).
Pattern	PPE-to-PPE flow: 2 ch�ques $\times$ 50,000 DT (round amounts, just under 100k threshold) from ALPHA corporate $\rightarrow$ Y personal. Immediate cash disposal. Reverse flow Y $\rightarrow$ ALPHA.
Predicate	Contrebande + corruption
CTAF outcome	Arrest of both PPEs + transmission to Procureur
Key signals	<code>is_ppe_to_ppe_flow</code> , <code>is_round_amount_under_threshold</code> , <code>is_immediate_cash_disposal</code> , <code>is_reverse_flow_to_recent_source</code>
Schema impact	PPE as binary flag layered on archetypes (D10). <code>controlled_accounts_list</code> (D15). <code>cheque_drawer_distribution</code> , <code>cash_disposal_velocity_*</code> , <code>reverse_flow_ratio_90d</code> .

5.4 Case 4 — Trade-Based Money Laundering (p. 22–23)

Actors	X (Franco-Tunisian, declared engineer, undisclosed g�rant-associ� of BETA — medical-dental import).
Pattern	KYC inconsistency: personal EUR account used as commercial conduit. Versements en esp�ces (138k EUR) + transfers from multiple physical persons (204k EUR) → transfers to foreign medical-dental suppliers (282k EUR). Sub-invoicing confirmed via TTN cross-check.
Predicate	TBML + customs fraud + tax fraud
CTAF outcome	Transmission to Procureur + judicial inquiry
Key signals	<code>is_kyc_inconsistency</code> , <code>is_personal_account_commercial_outflow</code> , <code>is_multi_physical_person_fan_in</code>
Schema impact	<code>declared_vs_observed_consistency_score</code> (KYC batch job, D16). <code>inflow_emitter_type_distribution</code> , <code>multi_physical_person_inflow_count_90d</code> . Counterparty features section 12 (D17). TTN integration deferred to production roadmap.

Observed CTAF severity ladder (non-monotonic — reflects procedural stage, not criminal severity): Case 1 (CNLCT listing) > Case 2 (asset freeze + prosecution) > Case 3 (arrest + prosecution) > Case 4 (transmission only).

6 Structural Extractions

6.1 Instruments Used in BA/FT Dossiers (p. 29)

Co-occurrence rates (a single dossier may use multiple instruments; percentages sum >100%):

Instrument	Rate
Esp�ces (cash)	68%
Virement	56%
D�livrance de fonds et mandats minute	49%
Ch�que	34%
Autres	13%

Mapping to our 4 operations: esp ces ↔ versement/retrait, virement ↔ virement, ch que ↔ remise de ch ques. D livrance de fonds and mandats minute are out of scope.

Generator calibration: anomalous sequences should be multi-operation (validated by co-occurrence >100%). When injecting multi-op anomalies, ~1 in 3 should involve a ch que in the chain.

6.2 Financial Tier Distribution (p. 14)

Enjeu financier of dossiers transmitted to Parquet (dossier-level cumulative amounts, not individual transactions):

Tier (TND)	2017	2018	2019	2020	2021
0–50k	2%	16%	11%	17%	3%
50k–100k	1%	2%	3%	2%	4%
100k–500k	6%	4%	8%	15%	27%
500k–1M	–	25%	6%	6%	9%
1M–10M	4%	22%	42%	30%	31%
>10M	87%	31%	30%	30%	26%

Key insight: 84% of 2021 dossiers exceed 100k TND cumulative. The near-empty low tiers (7% below 100k) likely reflect *detection bias* — existing rule-based systems surface large amounts more easily. Our system’s archetype-relative z-scores and LSTM sequence modelling are designed to close this gap.

Generator calibration: inject anomalies across all tiers. High-tier anomalies (>100k cumulative) are the “easy” cases; low-tier anomalies (<50k cumulative) should be the *hard* cases with subtler patterns and longer sequences, reflecting the real-world detection gap.

6.3 BBE Concentration Around Regulatory Thresholds (p. 38–39)

Interval (TND)	% Ops 2020	% Ops 2021	% Amount 2020	% Amount 2021
5–20k	87.57%	85.48%	51.21%	44.78%
20–50k	8.44%	9.30%	17.67%	17.77%
50–100k	2.71%	3.30%	12.74%	14.40%
100–200k	0.88%	1.50%	8.40%	12.91%
200–500k	0.32%	0.36%	6.68%	6.43%
>500k	0.07%	0.06%	3.30%	3.71%

Key insight: 85%+ of BBE operations cluster in the 5–20k TND band, straddling the 10,000 DT regulatory threshold. This is empirical evidence of threshold-aware structuring behaviour in Tunisia.

Schema validation: confirms the design of `near_threshold_ratio` (proportion of transactions in the 8,000–9,999 DT band) as a high-signal feature. Generator should produce heavy clustering in this band for smurfing-type anomalies.

6.4 Predicate Offence Taxonomy (p. 31)

See Section 4.2 above. Generator use: 3–4 scenario families with distinct operational fingerprints.

Sector cross-cut (personnes morales implicated in BA/FT dossiers):

Sector	% of implicated PM
Commerce international	50.71%
Services	17.14%
Nouvelles technologies	10.71%
Other sectors	21.44%

Design principle (D19): sector risk weight in scoring is *evidence-gated* — only CTAF-backed sectors receive non-zero weight; remaining sectors contribute zero to scoring but are retained for classification and dashboard filtering.

6.5 Geographic Concentration (p. 36–37)

BBE operations concentrated in 8 governorates over 2020–2021. Rank order (2021): Tunis (dominant), Médenine, Sousse, Ben Arous, Sfax, Nabeul, Ariana, Monastir.

Use: simulator realism calibration only. Branches in Tunis and Médenine see higher volumes and more anomaly injections in the live demo. No schema change.

6.6 Encodable Rules

Rule	Definition	Source	System location
R1	Client occasionnel making 3+ opérations à la file → reclassify as client habituel	CTAF signalement, 17 Sept 2021 (p. 24)	Batch profile rebuild: sets <code>regulatory_client_type</code> from <code>occasional</code> to <code>habitual</code> . This is a <i>regulatory classification</i> , distinct from <code>maturity_status</code> (statistical).
R2	Client occasionnel making 5+ versements for same beneficiary → potential passeur de fonds	CTAF BBE analysis, p. 27 (sustained across 2018–2021 reports)	Decision Service: regulatory rule. Enriched event already carries beneficiary tracking; Decision Service counts per beneficiary pair.

7 Sector Taxonomy — `accounts.sector_code`

18 coarse buckets. CTAF risk signal noted where evidence exists; all other sectors receive zero scoring weight (D19).

#	sector_code	Description	CTAF signal
1	<code>commerce.international</code>	Import/export trade	50.71% of implicated PM
2	<code>services.professional</code>	Legal, accounting, consulting	Part of “services” 17.14%
3	<code>technology</code>	IT, software, digital	“Nouvelles technologies” 10.71%
4	<code>commerce.retail</code>	Retail shops, local trade	–
5	<code>commerce.wholesale</code>	Wholesale, distribution	–
6	<code>construction.btp</code>	Construction, public works	–
7	<code>real.estate</code>	Property, development	Case 3: real estate transfer
8	<code>healthcare.medical</code>	Clinics, pharma, equipment	Case 4: medical-dental import
9	<code>tourism.hospitality</code>	Hotels, restaurants, travel	–
10	<code>agriculture.fishing</code>	Farming, livestock, fishing	–
11	<code>manufacturing</code>	Factories, production	–
12	<code>transport.logistics</code>	Freight, shipping, delivery	–
13	<code>financial.services</code>	Non-bank: leasing, micro-finance, insurance	–
14	<code>education</code>	Schools, training centres	–
15	<code>energy.extractive</code>	Oil, gas, mining	–
16	<code>telecom</code>	Telecoms, ISPs	–
17	<code>npo.association</code>	Non-profits, associations	Case 1: association as FT front
18	<code>other</code>	Catch-all	–

8 Schema Amendments (This Session)

8.1 Client Profile — New Field

Field	regulatory_client_type
Type	Enum: <code>occasional</code> <code>habitual</code>
Computed by	Nightly batch profile rebuild
Rule	If client has ≥ 3 operations $\rightarrow$ set <code>habitual</code> (irreversible)
Source	CTAF signalement 17 Sept 2021 (Rule R1)
Distinct from	<code>maturity_status</code> (statistical concept: enough data for reliable z-scores)

Combination semantics:

regulatory_client_type	maturity_status	Implication
occasional	cold	New client, 1–2 ops. Lighter KYC, archetype baselines as proxy.
habitual	cold	Crossed 3-op threshold: full vigilance, but insufficient history for reliable personal statistics. Cold-start defence layers active.
habitual	mature	Normal operating mode: full history, full vigilance.

9 Consolidated Architectural Decisions

Decisions D1–D18 were established in prior sessions (see weekly rapport Week 1). New decisions from this session:

#	Decision	Rationale
D19	Evidence-gated sector risk scoring	Only CTAF-backed sectors receive non-scoring weight. Remaining sectors contri zero but are retained for classification and fi ing. Do not pretend signal exists where evid is absent.
D20	<code>regulatory_client_type</code> distinct from <code>maturity_status</code>	Regulatory reclassification (3+ ops $\rightarrow$ habit and statistical maturity are independent di sions. A client can be regulatory-habitual statistically cold.
D21	Multi-tier anomaly difficulty gradient	High-tier anomalies (>100k cumulative) easy detection targets; low-tier anom (<50k) are hard cases with subtler patterns. flects the real-world detection bias revealed CTAF financial tier distribution.
D22	Multi-operation anomaly sequences	CTAF instrument co-occurrence rates (>1 sum) confirm BA/FT schemes are m instrument. Anomaly injection must gene multi-op sequences, not isolated events.

10 Generator Calibration Parameters (Consolidated)

All parameters derived from S3. To be combined with S1 (PaySim) and S2 (Amen Bank) parameters during generator design.

Parameter	Value	Source section
Anomaly scenario families	3–4 (cash-heavy, transfer-based, document-based)	Predicate taxonomy (§5.4)
Family weights	~20/19/11 (long tail remainder)	Predicate taxonomy
Multi-op sequence rate	Majority of anomalous sequences involve ≥ 2 operation types	Instrument co-occurrence
Chèque involvement	~1 in 3 anomalous sequences	34% instrument rate
Cumulative tier difficulty	Low tier (<50k) = hard; High tier (>100k) = easy	Financial tier distribution
Smurfing band clustering	Heavy concentration in 8,000–9,999 DT	BBE concentration (85% in 5–20k)
Geographic weighting	Tunis, Médenine, Sousse: higher anomaly density	Geographic cartography
Passeur threshold	5+ versements by same occasional client for same beneficiary	Encodable rule R2
BA/FT split	85% BA / 15% FT	CTAF transmission stats
CTAF-flagged sectors	commerce_international, services, technology, npo_association, real_estate, health_care_medical	Sector cross-cut + case studies

11 Pending Amendments from Prior Session

The following amendments were identified during CTAF case extraction (prior session) and remain pending formal application to Checkpoints 3.2–3.4.

11.1 Layer 1 — Raw Event

- `versement.payload.depositor_id`: enforcement strengthened (required when depositor $\neq$ holder)
- Verify: `remise_cheques.payload` exposes `drawer_account_type` and `drawer_account_id`
- Verify: `virement.payload` exposes `counterparty_country`

11.2 Layer 2 — Client Profile (new fields by source case)

- **Case 1:** `lifetime_versement_sum/count`, `lifetime_retrait_sum/count`, `lifetime_virement_in_sum` (domestic/international), `lifetime_third_party_deposit_count`, `lifetime_cash_conversion_ratio`

- **Case 2:** `account_count_active_90d/lifetime`, `multi_currency_activity_ratio_30d/180d`, `cross_jurisdiction_outflow_distribution`, `transit_velocity_p50/p95_30d/180d`
- **Case 3:** `controlled_accounts_list`, `cheque_drawer_distribution`, `cash_disposal_velocity_p50/p95_30d/180d`, `reverse_flow_ratio_90d`
- **Case 4:** `declared_vs_observed_consistency_score (batch)`, `inflow_emitter_type_distribution`, `outflow_destination_sector_distribution`, `multi_physical_person_inflow_count_90d`
- **This session:** `regulatory_client_type (occasional/habitual)`

11.3 Layer 3 — Enriched Event

New boolean rule signals (15 fields): `is_third_party_deposit`, `is_foreign_inbound_implausible_motive`, `is_high_velocity_transit`, `is_multi_currency_client`, `is_cross_jurisdiction_concentrated_outflow`, `is_ppe_to_ppe_flow`, `is_ppe_to_ppe_via_controlled_account`, `is_corporate_to_personal_same_controller`, `is_round_amount_under_threshold`, `is_immediate_cash_disposal`, `is_reverse_flow_to_recent_source`, `is_kyc_inconsistency`, `is_personal_account_commercial_outflow`, `is_multi_physical_person_fan_in`, `is_personal_foreign_currency_sector_concentration`. New section 12: counterparty-side derived features.

11.4 Batch Layer

- `archetype_baselines` segmented by sector $\times$ archetype $\times$ flag-set
- KYC consistency batch job (weekly cadence)
- `regulatory_client_type` reclassification (nightly cadence, 3-op threshold)

11.5 Cold Store

- `accounts` table: add `sector_code` (18-bucket enum)
- Verify `currency` and `account_type` columns exist

12 Next Milestones

Immediate:

1. Formally apply schema amendments to Checkpoints 3.2–3.4
2. S4 — BCT 2021 Annual Report extraction (v2 template); locate circulars 2018-09, 2017-08, 2021-05
3. Resolve D18: exhaustive regulatory threshold enumeration from BCT

Next build phase:

4. Scoring Service design (Autoencoder + LSTM coexistence, score fusion, SHAP co-location)
5. MLflow integration
6. Synthetic data generator implementation

Deferred / production roadmap:

7. TTN integration (cross-system TBML detection — Case 4)
8. Extension to 5th channel for investment products (Case 2)